A Quantitative Framework for Evaluating Explainable AI in Network Intrusion Detection Systems

Problem Statement

Modern Machine Learning-based Intrusion Detection Systems (IDS) achieve high detection accuracy but operate as **black boxes**. Network security analysts receive alerts without any justification, making it difficult to trust, triage, or investigate incidents. This lack of explainability leads to alert fatigue, missed attacks, and slow incident response—especially in environments generating tens of thousands of alerts daily.

While Explainable AI (XAI) methods such as SHAP and LIME exist, their application to network intrusion detection remains largely **ad-hoc**. There is no standardized framework to evaluate whether the explanations they produce are faithful, stable, or practically useful for the IDS domain.

Objective

Build an end-to-end IDS pipeline (classifier + explainer) and propose **three quantitative metrics** for evaluating explanation quality specifically on network traffic datasets. The goal is to establish a reproducible benchmark that enables fair comparison across XAI methods and drives adoption of explanations in production IDS.

Methodology

1. **Dataset Preparation**: Use CIC-IDS2017 and UNSW-NB15—labeled network flow datasets with 80+ features covering normal traffic and multiple attack types.

2. **Model Training**: Train Gradient Boosting and Random Forest classifiers (high performance, tree-based, natively compatible with SHAP).

3. **Explanation Generation**: Apply SHAP (game-theoretic) and LIME (local approximation) to produce per-prediction feature contribution values.

4. **Quantitative Evaluation**: Measure explanation quality along three axes:

Metric	What it measures
Fidelity	If top-k features are removed, does the prediction flip as expected?
Stability	Do similar inputs produce similar explanations?
Speed	Wall-clock time per explanation (real-time feasibility)

Expected Contributions

1. **First standardized evaluation** of XAI methods specifically designed for the network intrusion detection domain, addressing unique challenges such as class imbalance, temporal features, and high-dimensional categorical data.
2. **Open-source reproducible pipeline** (Python scikit-learn + shap + lime) that future researchers and practitioners can directly extend or benchmark against.
3. **Practical deployment recommendations**—a clear guide on which explainer to use under different latency and accuracy constraints (e.g., SHAP for offline analysis, LIME for real-time edge deployment).

Impact

- **For researchers**: A common evaluation baseline so future XAI-for-IDS papers are directly comparable, accelerating progress in the field.
- **For practitioners**: Evidence-based guidance on integrating explanations into production SOC workflows, reducing mean-time-to-respond (MTTR).
- **For the field**: A shift from "detect and alert" to "detect, explain, and justify" as the default IDS architecture.

Timeline

Weeks	Activity
1–2	Literature review + dataset preparation and cleaning
3–4	Model training + SHAP / LIME integration
5–6	Metric implementation + experimental evaluation
7	Paper writing and figure preparation
8	Submission to workshop / conference